Naga Venkata Sai Nadh Vaka

Master's student at OTH Regensburg seeking a Working Student Role

naga.vaka@st.oth-regensburg.de | Obertraubling, Germany (Flexible to commute or relocate)

(+49) 15510283549 | LinkedIn | GitHub | Engineering Portfolio | Available Immediately

Student Visa (140 Full Days / 280 Half Days per calendar year)



PROFILE

Master's student in Electrical and Microsystems Engineering at OTH Regensburg with a strong interest in semiconductor technology, digital IC design, and embedded systems. Currently expanding my knowledge of semiconductor memory and digital hardware design through hands-on projects, and eager to contribute to a collaborative engineering team while learning from experienced engineers.

EDUCATION

03/2026 – Present
Regensburg, Germany

Master of Engineering in Electrical and Microsystems Engineering,
Ostbayerische Technische Hochschule (OTH) Regensburg

- **Relevant Coursework:** Advanced Semiconductor Technology, Electronic Circuits & Systems, Electronic Product Engineering.

12/2021 – 04/2025
Vijayawada, India

Bachelor of Technology in Electronics and Communication Engineering,
Usha Rama College of Engineering and Technology

- **Relevant Coursework:** Digital IC Design, VLSI Design, Linear Integrated Circuits & Applications, Electronic Devices & Circuits, Electronic Circuit Analysis.
- **Academic Merit:** Graduated with Distinction (CGPA: 8.85/10 | German Grade Equivalent: 1.6)
- **Qualified GATE 2024** (Electronics & Communication Engineering), India.

TECHNICAL SKILLS

Hardware Description Languages: Verilog, SystemVerilog (Basic), VHDL

Programming: Python, C, MATLAB

Hardware & IC Design: RTL Design, Digital IC Design, CMOS Fundamentals, VLSI Fundamentals, Functional Verification, Embedded Systems, Microcontrollers

Laboratory: Oscilloscopes, Digital Multimeters, DC Power Supplies, PCB Prototyping

Simulation & Engineering Tools: AMD Vivado, Git, Keil µVision, Multisim, Proteus, Arduino IDE

Languages: English (C1 – Fluent), German (B1 – Actively Learning)

RELEVANT PROJECTS

05/2026 – Present

AXI4-Lite SPI Flash Controller | Verilog, AMD Vivado, *(In Progress)*

- **Designed** an AXI4-Lite register interface and FSM-based SPI controller enabling memory-mapped Flash access using Verilog.
- **Integrated** SPI transaction sequencing, register decoding, and Flash command execution into a synthesizable RTL design.
- **Verified** the design through custom Verilog testbenches and FPGA implementation, achieving **0 setup and hold timing violations** at a **100 MHz** target clock.
- **Optimized** hardware utilization to **193 LUTs** and **206 Flip-Flops** on a Xilinx Virtex-7 FPGA.

02/2023 – 05/2023

Digital Logic Design using Verilog & VHDL, *Bachelor's Laboratory Projects*

- **Developed** HDL implementations of combinational and sequential digital circuits using VHDL.
- **Implemented** multiplexers, encoders, decoders, flip-flops, counters, and shift registers with custom testbenches.

- **Verified** functionality through simulation and synthesized hardware schematics using Xilinx Vivado.

PUBLICATION

03/2025

Advanced Image Encryption Algorithm Integrating Chaotic Image Encryption and CNN, *i-manager's Journal on Electronics Engineering*

- **Published** a peer-reviewed journal article on a hybrid image encryption framework integrating chaotic cryptography and convolutional neural networks.

LEADERSHIP & TEAMWORK

08/2024 – 04/2025
Vijayawada, India

President — ECRATZ, ECE Student Association, *Usha Rama College of Engineering and Technology*

- Elected to represent **700+ engineering students** and coordinated technical events with faculty and student volunteers.

09/2022 – 04/2025
Vijayawada, India

Class Representative, *Usha Rama College of Engineering and Technology*

- **Coordinated** communication between faculty and a cohort of **60+ students**, ensuring timely dissemination of academic and administrative information.